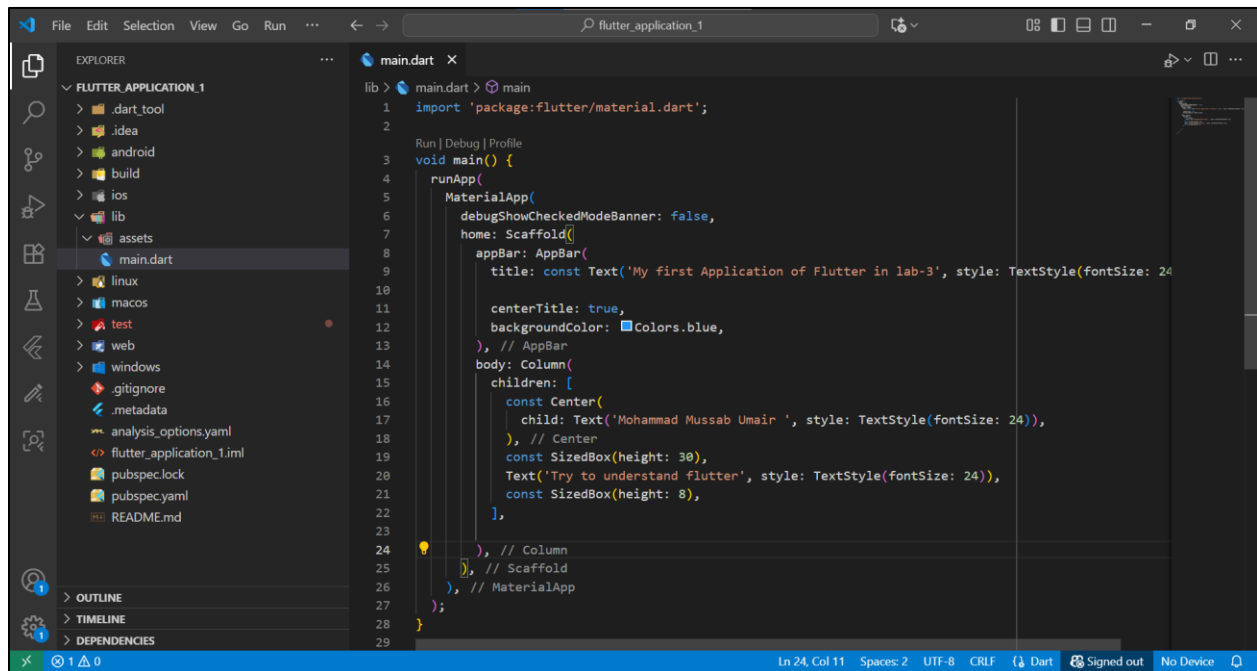


Lab Plan – 3: Flutter UI Basics and Widgets

Objective: The objective of this lab is to introduce students to the basics of Flutter, including project creation, widget usage, AppBar customization, UI design using rows and columns, handling spacing, and working with images (assets and network).

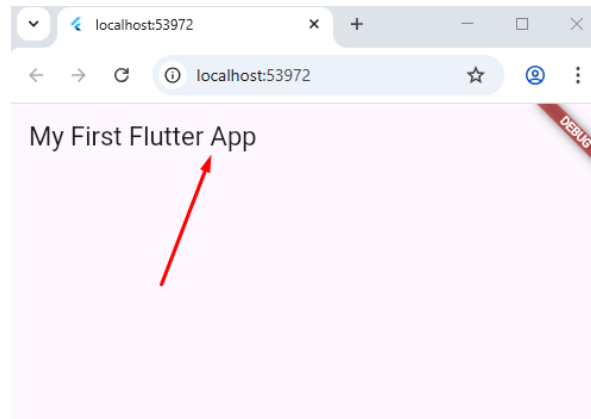
Task 1: Create a Flutter Project

- Open VS Code.
- Use shortcut:
 - **Ctrl + Shift + P** → Flutter: New Project
- Name your project and set up folder location.



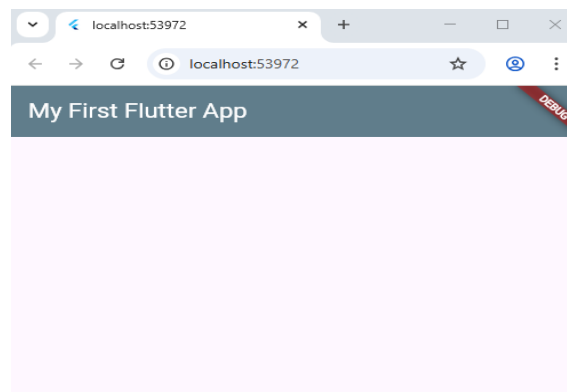
Task 2: Create a Stateless Widget and Add AppBar

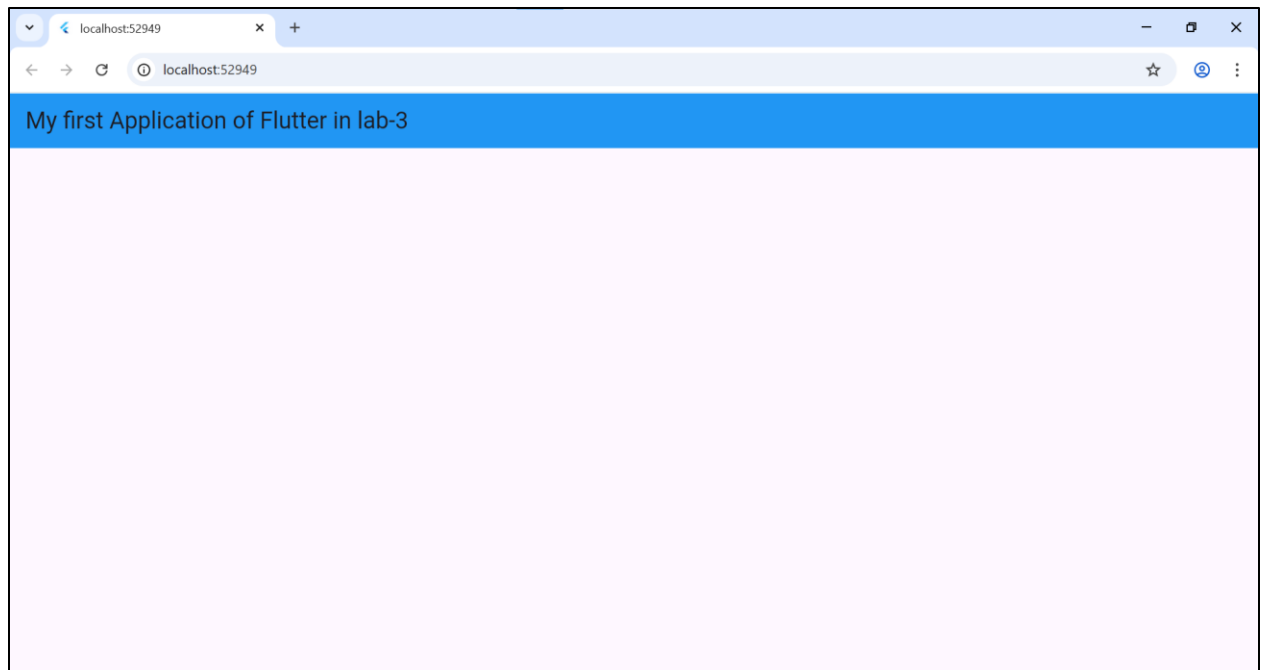
- Use shortcut **stle** to create a Stateless widget.
- Add an AppBar inside a Scaffold widget.



Task 3: Change the Color of AppBar

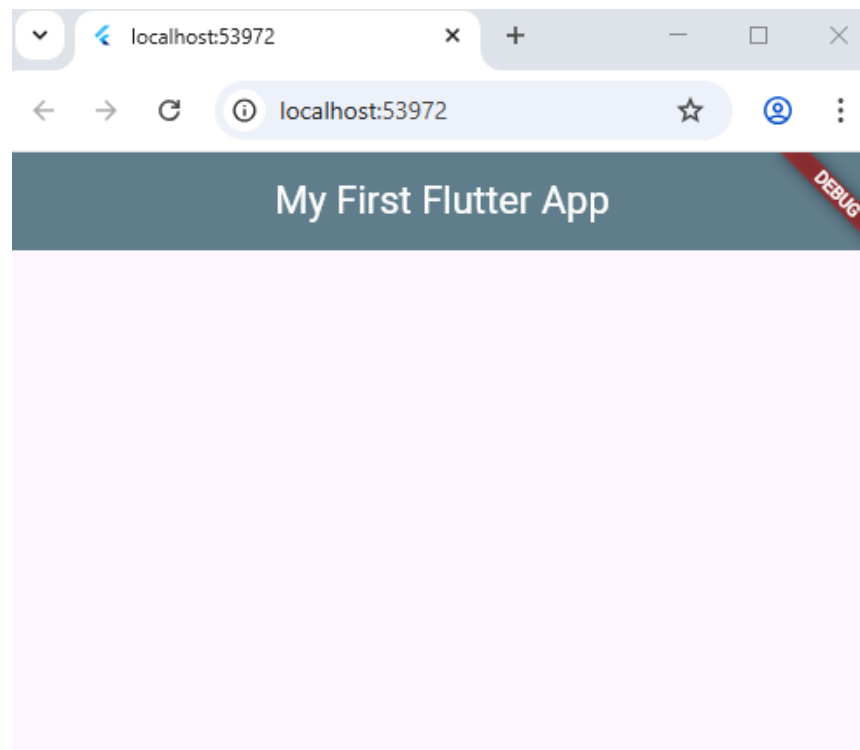
- Modify the **backgroundColor** property of AppBar.

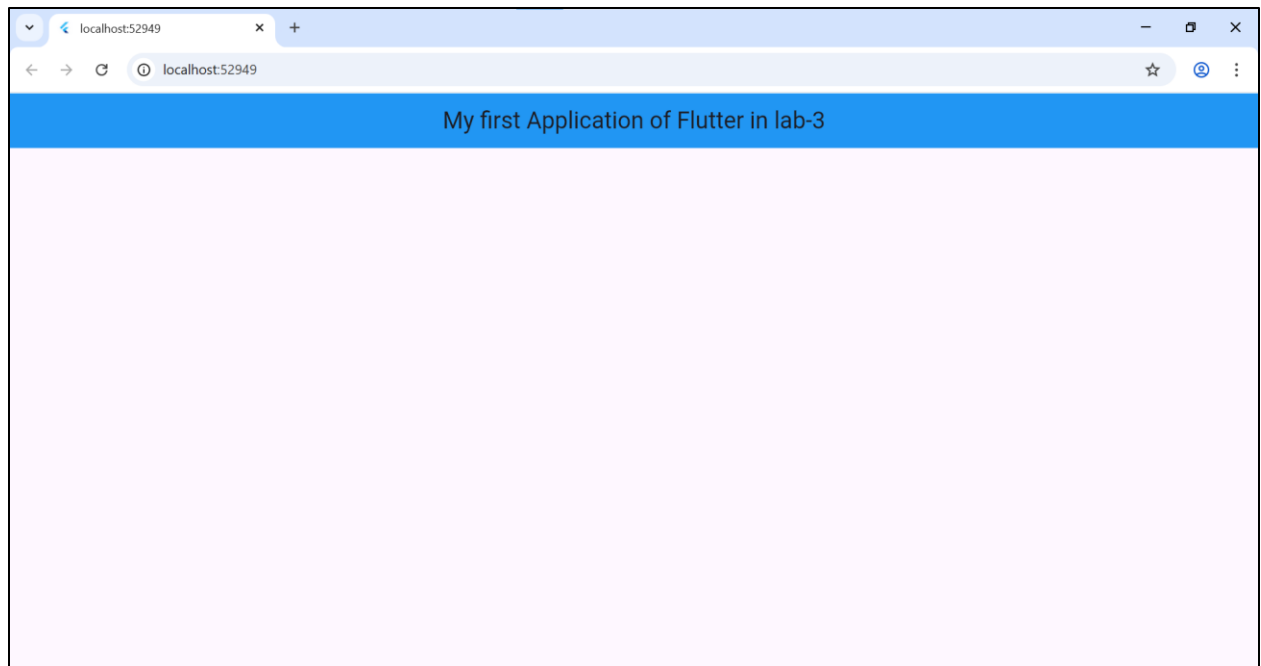




Task 4: Align the Text in AppBar to Center

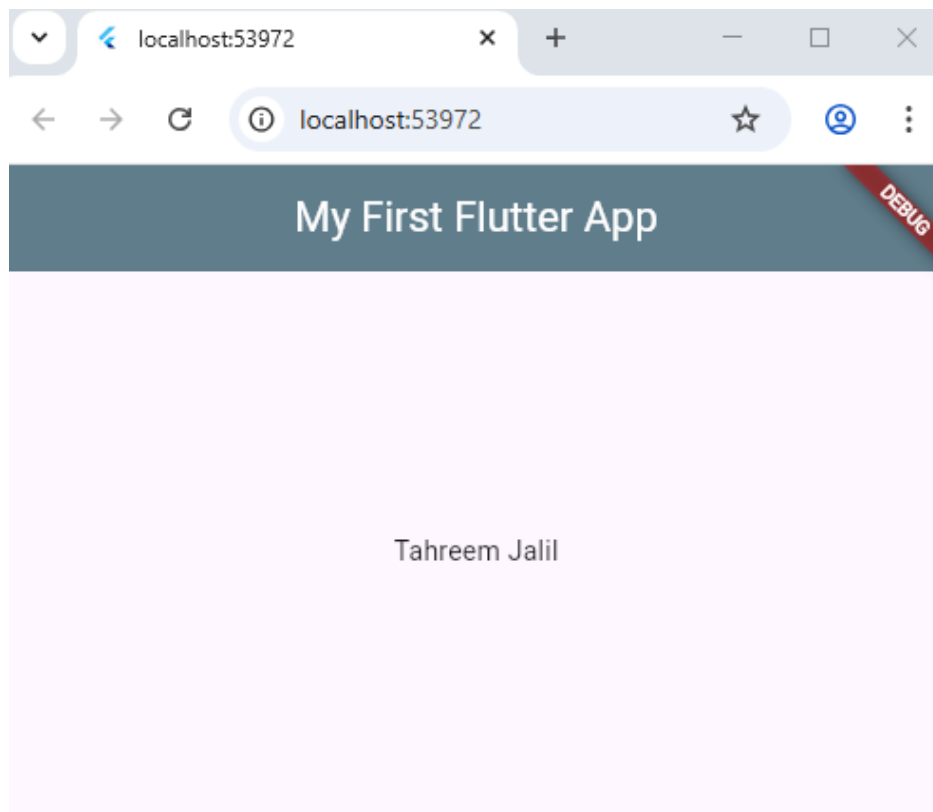
- Use **centerTitle: true** in AppBar.

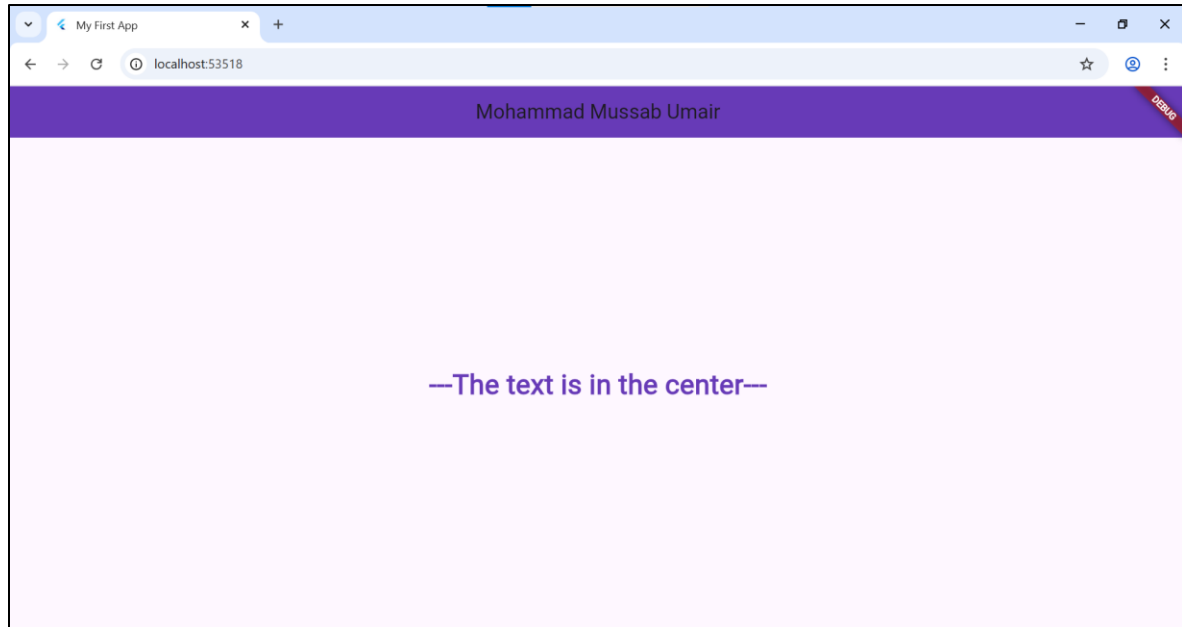




Task 5: Add Your Name at the Center of the Screen

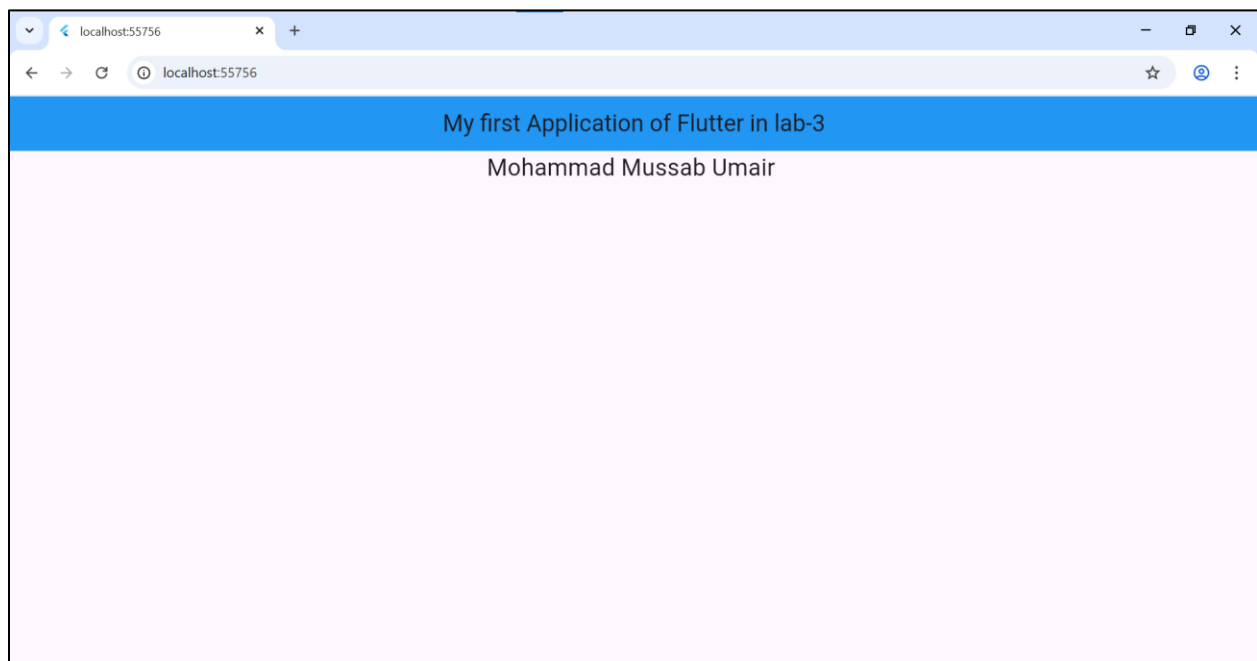
- Use the Center widget in the body section of Scaffold.





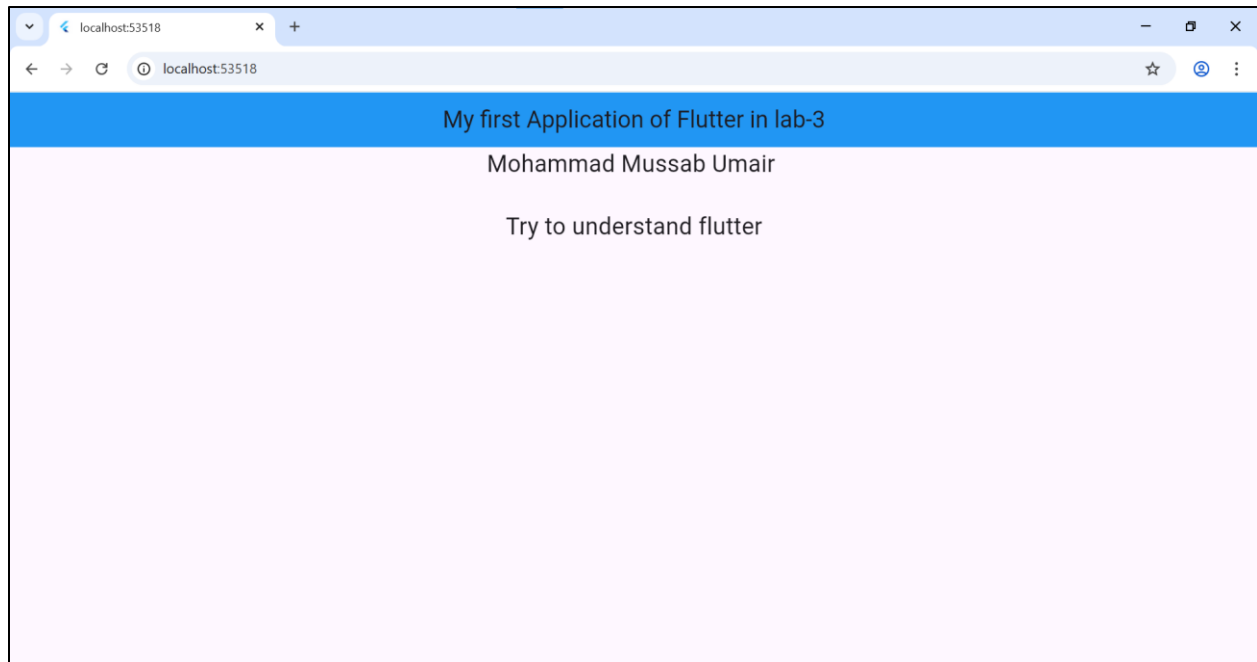
Task 6: Design the UI Layout

- Inside the Scaffold, create a Column widget.
- Add a Row widget as the first child of Column.



Task 7: Use SizedBox for Spaces

- Insert **SizedBox(height: --)** or **SizedBox(width: --)** to manage spacing between widgets.



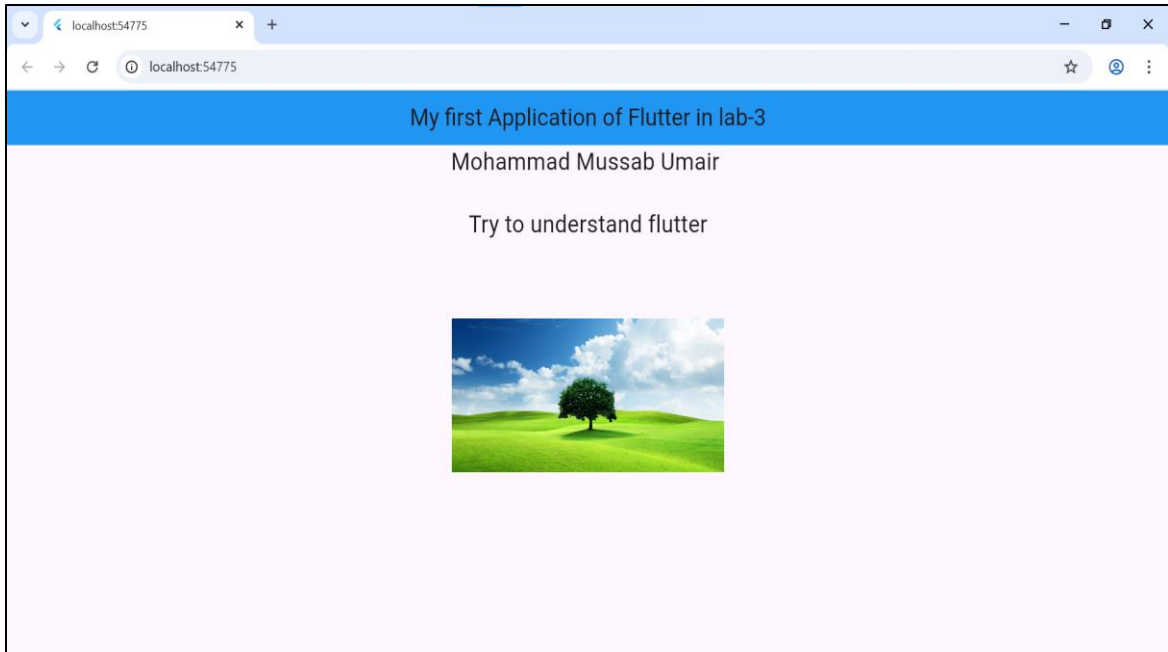
Task 8: Work with Images

(a) Add Assets Images

1. Download an image from Google.
2. Save it inside the **assets/images/** folder in your project.
3. Update the pubspec.yaml file: By using **Ctrl+S** or click **Pub get**

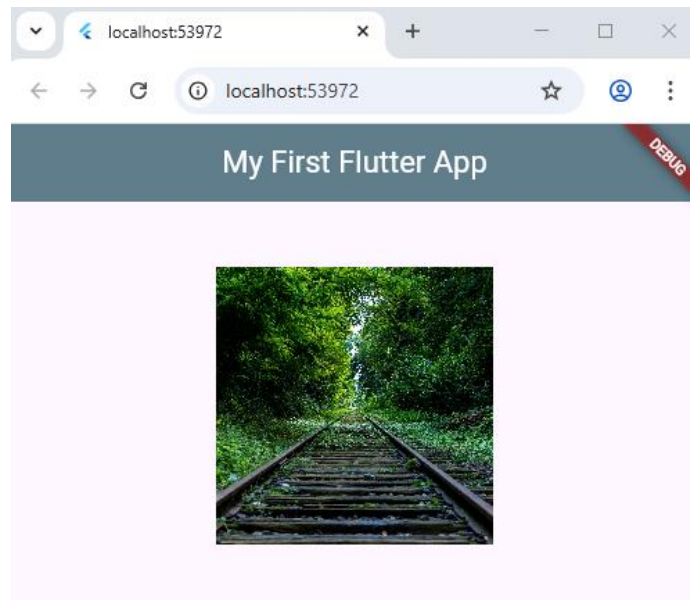
```
yaml

flutter:
  assets:
    - assets/images/
```

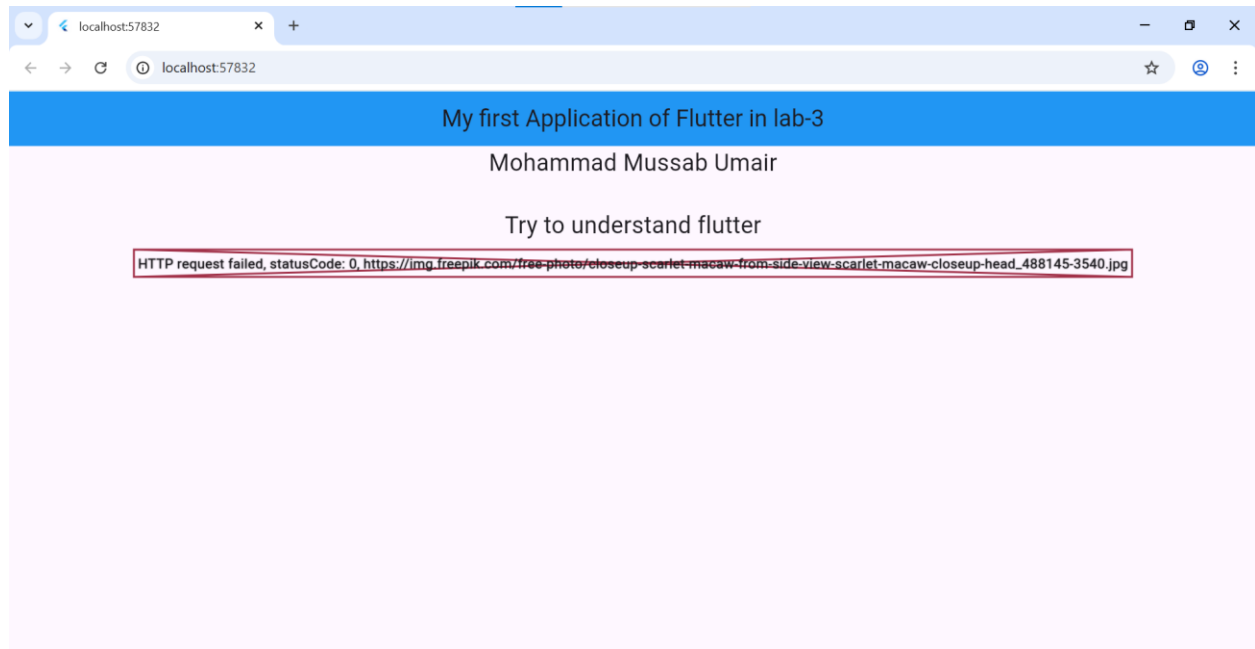


(b) Add Network Image

- Use `Image.network("URL")` to load image directly from the internet.



Error in this task so that why it shows error



Code:

```
import 'package:flutter/material.dart';

void main() {
  runApp(
    MaterialApp(
      debugShowCheckedModeBanner: false,
      home: Scaffold(
        appBar: AppBar(
          title: const Text(
            'My first Application of Flutter in lab-3',
            style: TextStyle(fontSize: 24),
          ),
          centerTitle: true,
          backgroundColor: Colors.blue,
        ),
        body: Column(
          children: [
            Row(
```



```
mainAxisAlignment: MainAxisAlignment.center,  
children: const [  
  Text(  
    'Mohammad Mussab Umair',  
    style: TextStyle(fontSize: 24),  
  ),  
],  
,  
const SizedBox(height: 30),  
const Text(  
  'Try to understand flutter',  
  style: TextStyle(fontSize: 24),  
,  
const SizedBox(height: 8),  
Image.asset(  
  'assets/images/tree-wallpaper.jpg',  
  height: 300,  
  width: 300,  
,  
],  
,  
,  
,  
,  
);  
}
```